

Mapping Affordable Housing Potential in Pittsburgh:
A Spatial Analysis of Housing Affordability and City-Owned Parcels in Pittsburgh
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Background

A significant portion of Pittsburgh's land remains vacant or underutilized, much of it held by the city itself, creating both challenges and opportunities for housing development. In fact, there are more than 8,000 properties owned by Pittsburgh or the Urban Redevelopment Authority, and these properties make up 9% of land within city limits.¹ This unused land does not contribute to the economy and actively harms surrounding communities. When parcels sit unused for years, they fall into disrepair, accumulate liens, and become increasingly difficult for outside organizations to redevelop.²

At the same time, Pittsburgh is experiencing the same housing pressures seen across the country. Although the city is often described as relatively affordable, many residents still spend more than 30% of their income on housing, and the supply of affordable homes remains low.³ The discrepancy between available land and community need widens existing inequities by making it harder for families to find stable, accessible housing.

This problem is further exacerbated given the scale of publicly owned land. According to the Western Pennsylvania Regional Data Center, the city currently holds **12,761** parcels in total.⁴ These parcels represent a largely untapped opportunity. If we could identify which parcels are ideal for redevelopment depending on need in that area, these parcels can be opportunities for new affordable housing development.

Project Proposal

The purpose of this project is to **identify priority areas for affordable rental housing development** in Pittsburgh by combining neighborhood-level rental affordability indicators with the spatial distribution of city-owned parcels available for sale.

¹ Zolkower, Justin. 2023

² Zolkower, Justin. 2023

³ Habitat for Humanity of Greater Pittsburgh. 2024

⁴ City of Pittsburgh. (2023). *City-Owned Properties Dataset*. WPRDC (Western Pennsylvania Regional Data Center).

This analysis integrates two key components:

1. Median gross rent and household income at the neighborhood level to measure housing affordability.
2. The location and density of City-owned parcels that are currently purchasable.

By overlaying socio-economic indicators with patterns of public land availability, the project highlights neighborhoods where redevelopment could have the greatest impact. In particular, the analysis seeks to identify areas with high concentrations of available parcels located in lower-income or lower-rent neighborhoods, where increasing the supply of affordable housing could most effectively support residents. Overall, identifying priority redevelopment areas is a critical first step toward revitalizing underused public land and expanding accessible housing options across the city.

The project focuses on the City of Pittsburgh. City-owned parcels will be filtered to include only those available for sale and aggregated to the neighborhood level to identify areas with the greatest need and opportunity for future affordable housing development.

Methodology

In this analysis, I will be using the following datasets:

1. ACS Median Household Income from US Census Bureau, 2023 ⁵
2. ACS Median Gross Rent from US Census Bureau, 2023 ⁶
3. City Owned Properties from the Western Pennsylvania Regional Data Center⁷

For the parcels, I am filtering for parcels that fulfill the following requirements:

- Class – Building or Vacant Land
- Inventory-Type – Public Sale
- Current Status: – Available for Sale
- Parcel Square feet > 0
- Zoning would include residential zones and mixed/commercial zones. Excluded parks, industrial zones, restricted areas

⁵ U.S. Census Bureau. (2023). *American Community Survey 5-Year Estimates (Table B19013: Median Household Income; Table B25064: Median Gross Rent)*.

⁶ U.S. Census Bureau. (2023). *American Community Survey 5-Year Estimates (Table B19013: Median Household Income; Table B25064: Median Gross Rent)*.

⁷ City of Pittsburgh. (2023). *City-Owned Properties Dataset*. WPRDC (Western Pennsylvania Regional Data Center).

I implemented three different analyses to determine where publicly owned parcels that may be suitable for potential affordable housing intersect with economic need in Pittsburgh.

- Kernel Density of Available Parcels – I used a Kernel Density Estimation (KDE) to visualize concentration of city-owned parcels. The KDE produced a continuous density surface classified into five levels from Very Low to Very High Density of parcels. This is measured by points (parcels) per square foot within the search area. In other words, this map shows where the highest clusters of potentially developable city owned parcels are located.
- Medium Household Income with Parcel Aggregation – I used the American Community Survey (ACS 2023) block group data to map the median household income across Pittsburgh. I created a choropleth using a green color. The parcel data was spatially joined to block groups, summarized, and aggregated to neighborhood level. Neighborhood parcel totals were then overlaid as proportional yellow circles. This map shows how parcel availability correlates with economic vulnerability, highlighting neighborhoods with both low incomes and high quantities of public land.
- Gross Rent with Parcel Aggregation – I used the median gross rent from ACS 2023 to map a choropleth using a blue gradient to represent cost burden. I used the same proportional parcel count symbols to overlap on this map to see if high quantity of public land coincides with high or low housing costs.

Results

Using this methodology, I was able to generate the following maps that helped inform my analysis of areas that would be ideal for affordable housing organizations. The kernel density map, highlighted in *Figure 1*, reveals that available city-owned parcels are not evenly distributed across Pittsburgh. Instead, they form **three distinct very high-density clusters**, all located in neighborhoods that have historically faced challenges related to disinvestment, lower housing quality, and economic instability.

Figure 1: Spatial Density of City-Owned Properties in Pittsburgh is calculated by parcels per square foot. Darker clusters means there is a high density of parcels.

1. **Homewood** Cluster – Very High

Density of Available Parcels

The strongest concentration of available parcels throughout Pittsburgh occurs across Homewood West, Homewood North, and Homewood South, extending toward East Hills, and even extending to the Lincoln-Lemington-Belmar region. These neighborhoods have historically been affected by housing vacancy and structural disinvestment. This explains the large volume of publicly owned parcels available for redevelopment.

2. The second high cluster of parcels in between the neighborhoods Garfield and **Central Lawrenceville**. Central Lawrenceville has a total parcel of 163, contributing to the large cluster in this area.

3. The last high cluster density is in the neighborhood **Beltzhoover**.

Spatial Density of City-Owned Properties in Pittsburgh

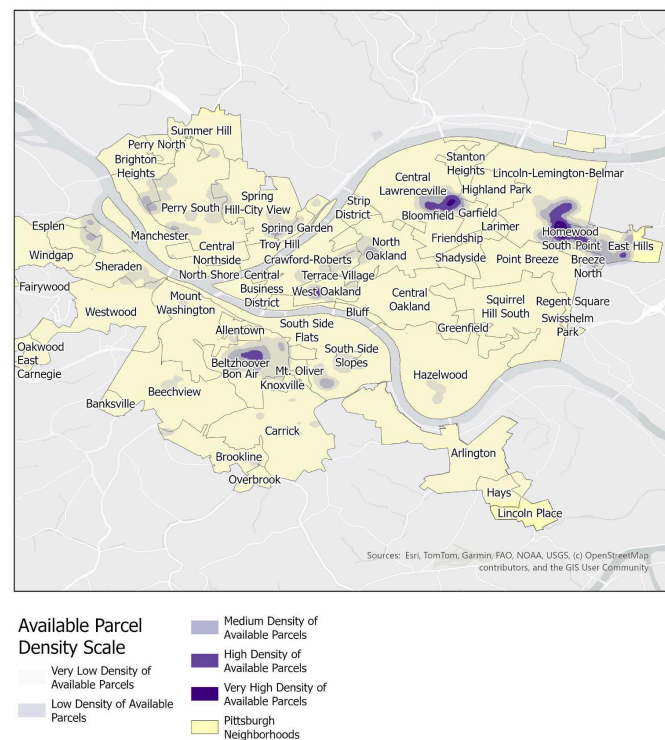


Figure 1

In addition, there are also several clusters of medium density areas in Pittsburgh. For example, the neighborhoods include South Side Slopes, Hazelwood, North Oakland, and a few others that are scattered around Pittsburgh. While these areas contain fewer parcels than the major hotspots, the

presence of medium-density clusters still represents meaningful redevelopment potential. These areas can be useful for small-scale affordable housing initiatives.

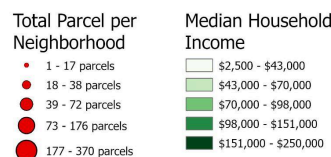
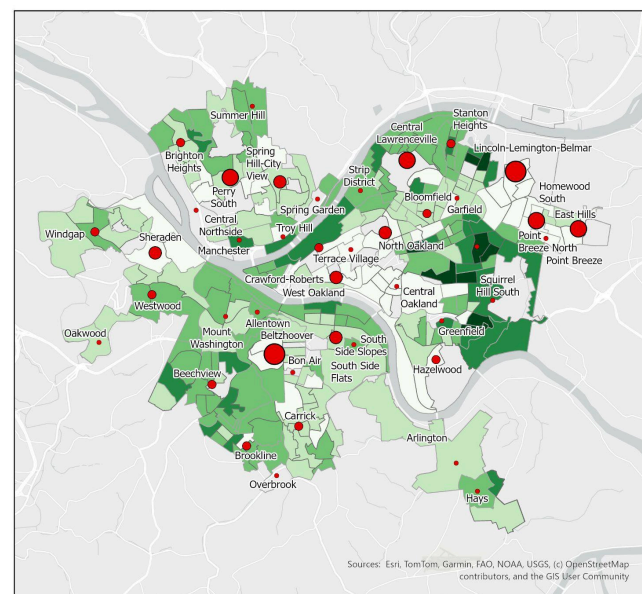
The key insight from this map shows large portions of Pittsburgh, such as Shadyside, Squirrel Hill, South Side Flats, Lawrenceville, and the Strip District, show no visible density of available city-owned parcels. Therefore, the majority of Pittsburgh's neighborhoods show minimal opportunity for development using publicly owned land, making the high density areas that the map identified especially key areas for housing policy and redevelopment planning.

Choropleth Maps

When examining *Figure 2*, it is evident that low-income areas have the highest amount of available parcels from the city. Similar to Figure 1, Homewood, East Hill, and Belthoover have high amounts of available parcels owned by the city and fall into the lowest income brackets. By summing all parcels located within the lowest-income block groups, I calculated that **1,113 out of 1,890** available parcels fall into this category. In other words, **approximately 59% of all available parcels are located in the lowest-income areas of the city**, indicating that the vast majority of public land suitable for housing redevelopment is concentrated in neighborhoods with high economic need.

In contrast, high-income neighborhoods such as Squirrel Hill, Shadyside, Point Breeze, have almost no city-owned parcels available for redevelopment. Mid-income areas display scattered availability rather than concentrated clusters. This pattern highlights that the city's capacity to create new housing through publicly controlled land is overwhelmingly located in historically disinvested and generally low-income neighborhoods. Some block groups lack ACS income data (such as portions of Central Lawrenceville), but the overall pattern remains clear: redevelopment potential through

Median Household Income and Neighborhood Parcel Availability in Pittsburgh (2023)



public land is closely related with areas of greatest economic need. In simple terms, the places that need the most help are also where the city has the most land to rebuild/redevelop.

To better understand how available parcels are distributed across Pittsburgh, I created a neighborhood-level layer by aggregating parcel counts from all block groups within each neighborhood. This revealed substantial variation in parcel availability. **Lincoln-Lemington** has the highest number of available parcels (370), followed by **Beltzhoover** (243) and **East Hills** (176). Across the city, neighborhoods have an average of **45 parcels**, but the distribution is highly right-skewed. This means that a few neighborhoods hold a disproportionately large share of the available public land, which inflates the mean. Figure 3 below shows this distribution in parcels by neighborhood.

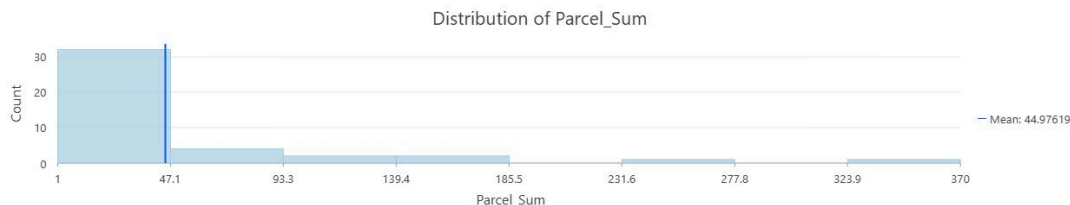
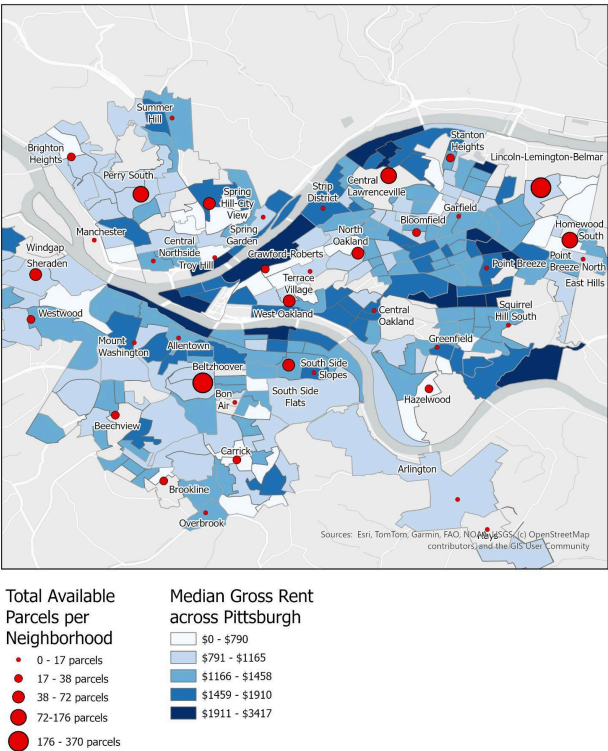


Figure 3

Lastly, the gross rent levels map (Figure 4) shows patterns similar to the income map. Darker blue areas represent higher rents, which helps indicate where renters may face higher cost burdens. Neighborhoods with the lowest rents, such as Lincoln-Lemington-Belmar and Beltzhoover, also have low incomes and some of the highest numbers of available parcels. Several neighborhoods, including East Hills, Homewood South, and Central Lawrenceville, do not have ACS rent data and appear blank on the map. Overall, Figure 4 shows that areas with the highest rents tend to have very few or no available parcels, while areas with lower rents have the most parcels available.

Gross Rent Levels and Available City-Owned Parcels in Pittsburgh



When comparing the income and rent maps together, a few key insights should be noted. Most high-rent areas are not low-income, and many low-income neighborhoods, such as Beltzhoover and Lincoln-Lemington-Belmar, actually have low to moderate rents. This suggests renters in these areas struggle more because of low income than high rent. These same neighborhoods also contain the highest number of available parcels. Overall, this reinforces that Pittsburgh's redevelopment opportunities are mainly in historically disinvested, low-income neighborhoods, while renters in high-rent areas face affordability pressures in places where the city has little or no land to develop new housing.

Conclusion/Future Work

Based on the combined analyses of income, gross rent, and parcel availability, the neighborhoods with the strongest potential for affordable housing development include Homewood (North, South, and West), East Hills, Beltzhoover, and Lincoln-Lemington-Belmar. These areas have both the highest concentrations of city-owned parcels and some of the lowest incomes in the city, showing clear need and strong redevelopment potential. Because rents in these neighborhoods are relatively moderate, new affordable units could realistically serve current residents and support long-term neighborhood stability.

Future work could include adding more neighborhood-level indicators, such as housing quality, vacancy, transit access, and proximity to amenities, to build a more detailed housing need score. These additions could further refine where affordable housing investments would be most effective.

Data Sources

U.S. Census Bureau. (2023). *American Community Survey 5-Year Estimates (Table B19013: Median Household Income; Table B25064: Median Gross Rent)*.

U.S. Census Bureau. (2023). *TIGER/Line Shapefiles: Census Block Groups*.

City of Pittsburgh. (2023). *City-Owned Properties Dataset*. WPRDC (Western Pennsylvania Regional Data Center).

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References

Zolkower, Justin. 2023. “*A Huge Problem: Thousands of City and URA Properties Stuck in Limbo, off the Market and Tax Rolls.*” *Pittsburgh Union Progress*, August 13, 2023.

Habitat for Humanity of Greater Pittsburgh. 2024. “Why Housing Matters.”